DATASCI 406 FA 26, Quiz 3.1

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

A bootstrap SE for a sample of $n = 30$ observations is computed from $B = 1000$ bootstrap samples. What changes if B is raised to 100,000?

1. The bootstrap SE converges to the true SE of T.
2. Both the Monte Carlo error and the error from using $\hat{F}$ in place of F shrink.
3. The Monte Carlo error shrinks, but the error from using $\hat{F}$ in place of F does not.
4. Nothing, because every one of the 30^{30} possible bootstrap samples is already drawn.

Solution

Answer: 3. B controls only how well the T_b^* describe the bootstrap distribution, which is the sampling distribution of T under $\hat{F}$. That distribution differs from the one under F no matter how large B is, so option 1 and option 2 are wrong. Option 4 is wrong because $30^{30} \approx 2 \times 10^{44}$, far more than 1000.

Question 2 (1 point)

When would you use a basic or percentile interval instead of the normal interval, $T \pm z_{1-\alpha/2}\,\widehat{\mathrm{SE}}$?

1. When the bootstrap distribution of T^* is skewed.
2. When B is small, because quantiles need fewer bootstrap samples than an SE.
3. When T is the mean of a large sample.
4. When the bootstrap SE is large.

Solution

Answer: 1. The normal interval uses only the SE, so it is symmetric about T and assumes T is approximately normal. The basic and percentile intervals use quantiles of the bootstrap distribution, so they can reach further on one side, as for the Flint 90th percentile without the two excluded samples. Option 2 is backwards: tail quantiles need a larger B than an SE does. Option 3 is the case where the normal interval works well. Option 4 is wrong because a wide symmetric distribution is still fit by the normal interval.

Question 3 (1 point)

Each of `n` homes has a first-draw lead level `x[k]` and a level after flushing `y[k]`. Which code computes one bootstrap replicate of the mean difference of logs?

1. `mean(log(sample(x, replace = TRUE)) - log(sample(y, replace = TRUE)))`
2. `i <- sample(n); mean(log(x[i]) - log(y[i]))`
3. `i <- sample(n, replace = TRUE); mean(log(x[i])) - mean(log(y))`
4. `i <- sample(n, replace = TRUE); mean(log(x[i]) - log(y[i]))`

Solution

Answer: 4. $\hat{F}$ puts mass $1/n$ on each pair, so resample rows and keep each home's two values together. Option 1 resamples the columns separately and breaks the pairing. Option 2 samples without replacement, which only reorders the homes and returns the observed mean every time. Option 3 resamples `x` but not `y`.

Question 4 (1 point)

For which of these statistics would the bootstrap **not** give a good estimate of the sampling distribution?

1. The sample correlation.
2. The sample maximum.
3. The sample median.
4. The mean difference of paired observations.

Solution

Answer: 2. A bootstrap sample can only reuse observed values, so T^* is never larger than the observed maximum, while the true maximum can be. The bootstrap needs a small change in F to make a small change in T. The correlation and the mean difference are smooth functions of means, so they work well. The median and other quantiles work roughly, as the Flint 90th percentile did.

Rubric. 1 pt per question, no partial credit.